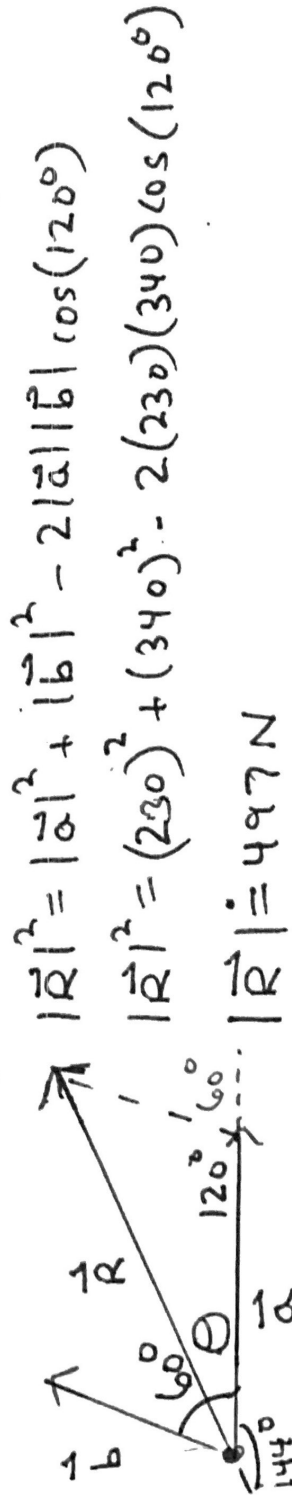


Example #2: Two draft horses pull a load. The chain between the horses and the load are at an angle of 60 degrees to each other. One horse pulls with a force of 230N and the other with a force of 340N.

(a) What is the resultant force on the load? In what direction is this force?

Let $\vec{a}$ and $\vec{b}$ represent the 230N and 340N forces respectively.

Let $\vec{R}$ represent the resultant force.



(b) What is the equilibrant force on the load? In what direction is this force?

The equilibrant force is 497N acting 144° to the 230N force.

$$180^\circ - 36^\circ = 144^\circ$$

Direction

$$\frac{\sin(\theta)}{340} = \frac{\sin(120^\circ)}{497}$$

$$\theta \doteq 36^\circ \text{ to}$$

the 230N force

Example #3: A traffic sign with a mass of 5kg is suspended above the street as shown below. Find the tensions (forces) in each of the wires.